

# Zeros, Multiplicity, and Graph Behavior

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*Zeros from factored form, factoring to find zeros, multiplicity and graph behavior, and building a polynomial from its zeros*

## Directions

- Show the work that produces each answer, then write the final answer on the blank provided.
- “Multiplicity” means the exponent on a factor: in  $(x - r)^m$  the zero  $x = r$  has multiplicity  $m$ .
- A zero of odd multiplicity is a place where the curve passes through the  $x$ -axis; a zero of even multiplicity is a place where the curve meets the axis and turns back.
- Every model in this set states its units. Answers to modelling questions should be given in those units.

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**1.** A lemonade stand’s daily profit, in dollars, is modelled by  $P(x) = (x - 2)(x - 8)$ , where  $x$  is the price charged per cup, in dollars. List the two prices at which the stand breaks even, that is, the prices where the profit is zero.

**Answer:** \_\_\_\_\_ dollars

**2.** The polynomial  $g(x) = (x + 3)^2(x - 1)$  is graphed on a coordinate plane.

(a) List all real zeros of  $g$ : \_\_\_\_\_

(b) One of those zeros is a place where the curve meets the  $x$ -axis and turns back instead of passing through it. Name that zero, and explain what feature of the factored form tells you it behaves that way.

**3.** An open tray is folded from a sheet of card. Its volume, in cubic centimetres, is  $V(x) = x^3 - 9x^2 + 20x$ , where  $x$  is the fold height in centimetres.

(a) Factor  $V(x)$  completely: \_\_\_\_\_

(b) List every value of  $x$  at which the volume is zero: \_\_\_\_\_

**4.** Let  $f(x) = (x - 1)(x + 4)^2$ .

(a) List all real zeros of  $f$ : \_\_\_\_\_

(b) Find the  $y$ -intercept,  $f(0)$ : \_\_\_\_\_

**5.** Let  $p(x) = -(x + 2)(x - 3)^2$ .

(a) Find  $p(1)$ : \_\_\_\_\_

(b) Describe what the graph of  $p$  does far to the right and far to the left, and describe what it does at  $x = 3$ . Give the reason in each case.

**6.** Consider  $q(x) = x^4 - 13x^2 + 36$ .

(a) Factor  $q(x)$  completely: \_\_\_\_\_

(b) List all real zeros of  $q$ : \_\_\_\_\_

**7.** A designer needs a degree-3 polynomial model whose graph meets the  $x$ -axis at  $x = -1$  and turns back there, crosses the  $x$ -axis at  $x = 4$ , and has leading coefficient 1.

(a) Write the model in factored form, then list its zeros: \_\_\_\_\_

(b) Multiply the model out into standard form: \_\_\_\_\_

**8.** A production cost, in dollars, is modelled by  $C(x) = a(x - 1)(x - 4)(x - 6)$ , where  $x$  is the number of units made, in hundreds, and  $a$  is a constant still to be chosen. The model must give a start-up cost of  $C(0) = 48$  dollars. Find the value of  $a$ .

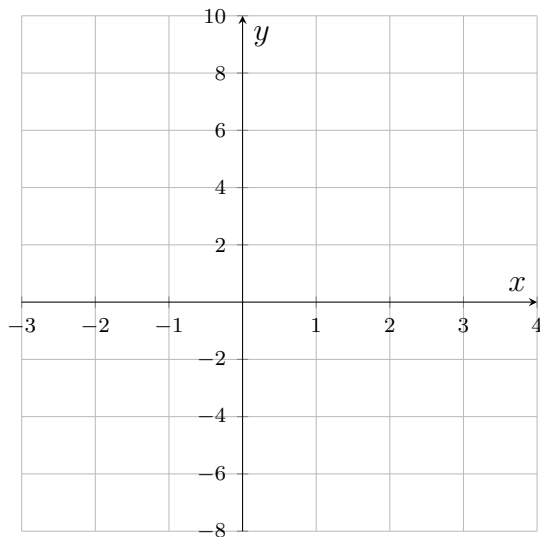
Answer: \_\_\_\_\_

9. Let  $f(x) = (x + 1)(x - 2)^2$ .

(a) List all real zeros of  $f$ : \_\_\_\_\_

(b) Find  $f(1)$ : \_\_\_\_\_

(c) Sketch the graph of  $f$  on the grid below. Make the curve show the correct behaviour at each zero and pass through the point you found in part (b).



**10.** A landscaped berm runs beside a footpath. Its height above the path, in metres, is modelled by  $R(x) = x^3 - 4x^2 + 4x$ , where  $x$  is the distance along the path from the start of the berm, in metres, for  $0 \leq x \leq 4$ .

(a) Factor  $R(x)$  completely: \_\_\_\_\_

(b) List the values of  $x$  at which the berm height is zero: \_\_\_\_\_

(c) One of those zeros is repeated. Explain what the repeated zero means for the shape of the berm where it meets the path.